

ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK

ACEnet: Free Cluster Computing For Researchers

Dr. Ross Dickson, Computational Research Consultant

What is ACEnet?

- “Atlantic Computational Excellence Network”
- Shared regional resource for ...
 - ... high-performance computing
 - ... remote collaboration
 - ... visualization
- Hardware, software, and personnel

Where is ACEnet?



Member institutions



Mount
Allison
UNIVERSITY



CAPE BRETON
UNIVERSITY

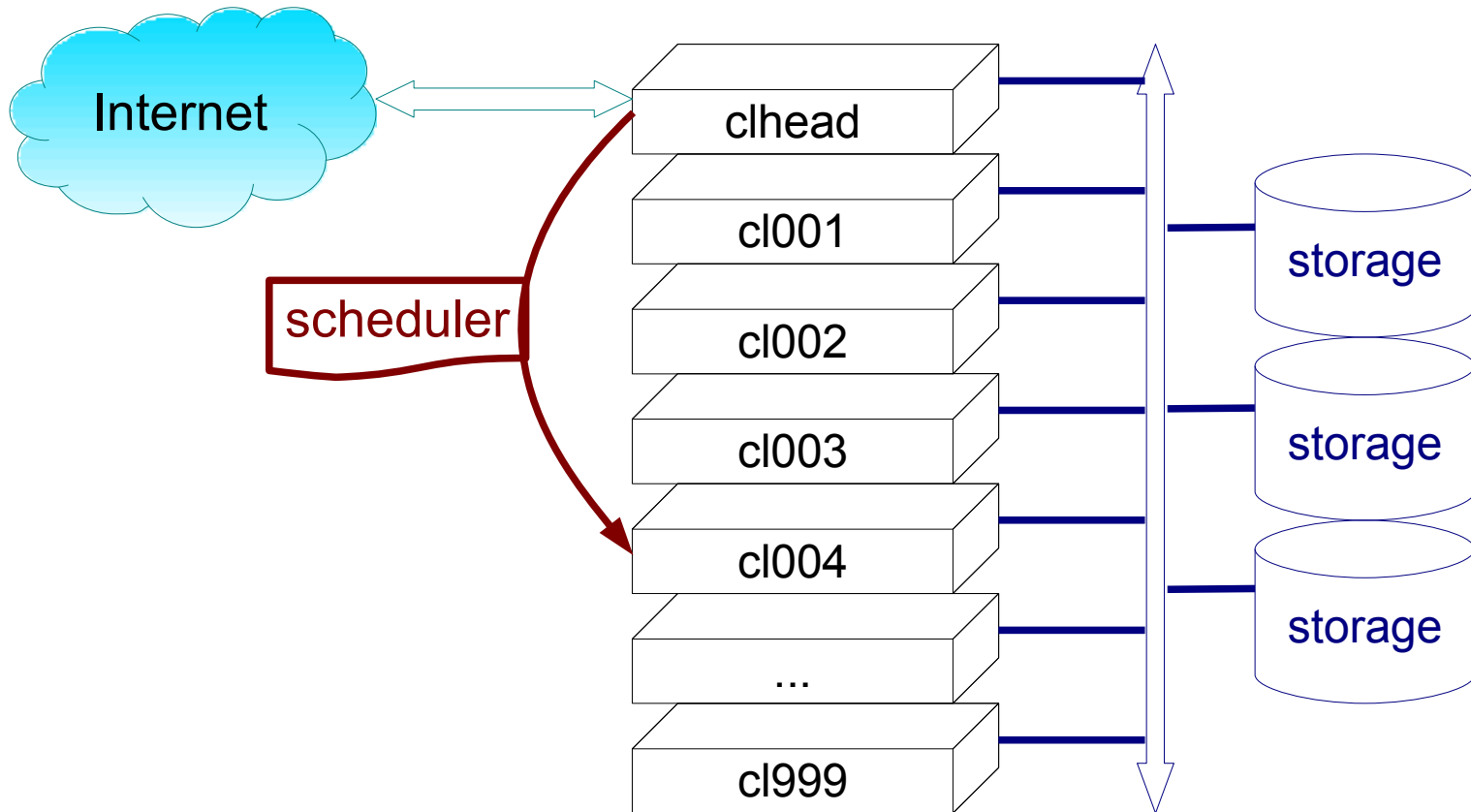


What's “High Performance Computing”?

- Many CPUs working on one problem
 - Many “serial” jobs running at once, or
 - many CPUs working in close coordination.
 - Communication between CPUs is key!
- Advantages
 - Faster results
 - Larger problems or more detailed simulations
- Principal tool: Computer **cluster**
 - Racks of regular computers, networked



Cluster workings



ACEnet Clusters

CPU and Memory Configurations



Cluster	Nodes	Cores/node	RAM/node	Total cores
Brasdor	269	4	8G	1152
	38	2	4G	
Courtenay	13	4	16G	52
Fundy	38	16	64G, 128G	636
	7	4	16G	
Glooscap	56	4	4G	848
	39	16	32G	
Mahone	139	4	16G, 64G	556
Placentia	154	16	32G, 64G	2884
	17	12	24G	
	53	4	16G, 64G	
	115	owned by individual groups		860

ACEnet Clusters

Storage Configurations



Cluster	/home user quota	/globalscratch user quota	/nqs size	/scratch/tmp
Brasdor	47 Gb	238 Gb	14 Tb	20 - 330 Gb
Courtenay	--	--	2.0 Tb	100 Gb
Fundy	47 Gb	238 Gb	14 Tb	35 - 100 Gb
Glooscap	13 Gb	48 Gb	2.8 Tb	0 - 15 Gb
Mahone	13 Gb	238 Gb	14 Tb	50 - 100 Gb
Placentia	13 Gb	238 Gb	14 Tb	0 - 140 Gb
Cluster-local				Node-local

Applications

https://wiki.ace-net.ca/index.php/Available_Software

<u>Molecular</u>	<u>Math&CS</u>	<u>Engineering</u>	<u>Earth &Ocean</u>	<u>Bio</u>
Gaussian	spin	ANSYS		mpiBlast
Gromacs	GAP	Fluent/	CDO	MrBayes
CPMD	DiVinE	Gambit	GMT	IM
Abinit	Macaulay-2	CFX	MM5	Migrate-n
NAMD	...	LS-DYNA		PhyML
WebMO		elmer-fem	<u>Astro</u>	PLINK
VMD			StarLab	
...			IRAF	

Software requests to:

support@ace-net.ca

Software Development

https://wiki.ace-net.ca/index.php/Available_Software

<u>Languages</u>	<u>Suites</u>	<u>Parallel</u> <u>APIs</u>	<u>Libraries</u>	<u>Tools</u>
C/C++	PGI		LAPACK	TotalView
Fortran	Intel	Open MPI	FFTW	dbx, gdb
Perl	Gnu	OpenMP	GSL	Valgrind
Python	SunStudio	...	ACML	make
Java			netCDF	cvs, svn
...			HDF	git
			Boost	...
			...	

Software requests to:

support@ace-net.ca

Data Analysis



https://wiki.ace-net.ca/index.php/Available_Software

R

Rmpi

Mathematica

Matlab*

*Distributed Computing Server
Octave

Python

Numpy

SciPy

Sage

pyMPI

pylab

matplotlib

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Remote Collaboration



Internet videoconferencing - inSORS & Access Grid

Dalhousie - Memorial - Mount Allison - Saint Mary's - Saint Francis Xavier
Acadia - New Brunswick + **other sites across Canada & the world**

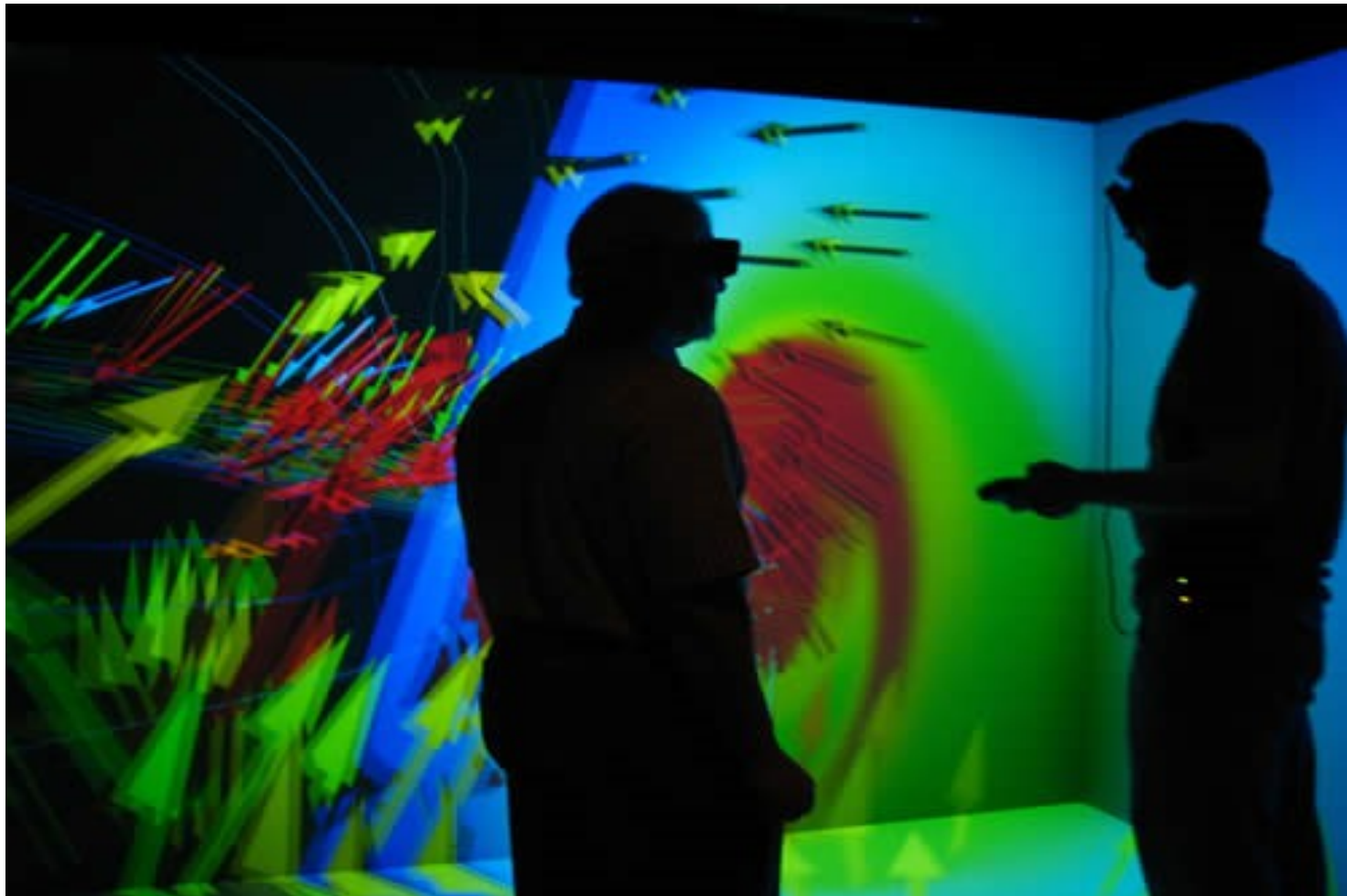


Advanced Visualization



Data cave

3D immersive data exploration facility @ Saint Mary's



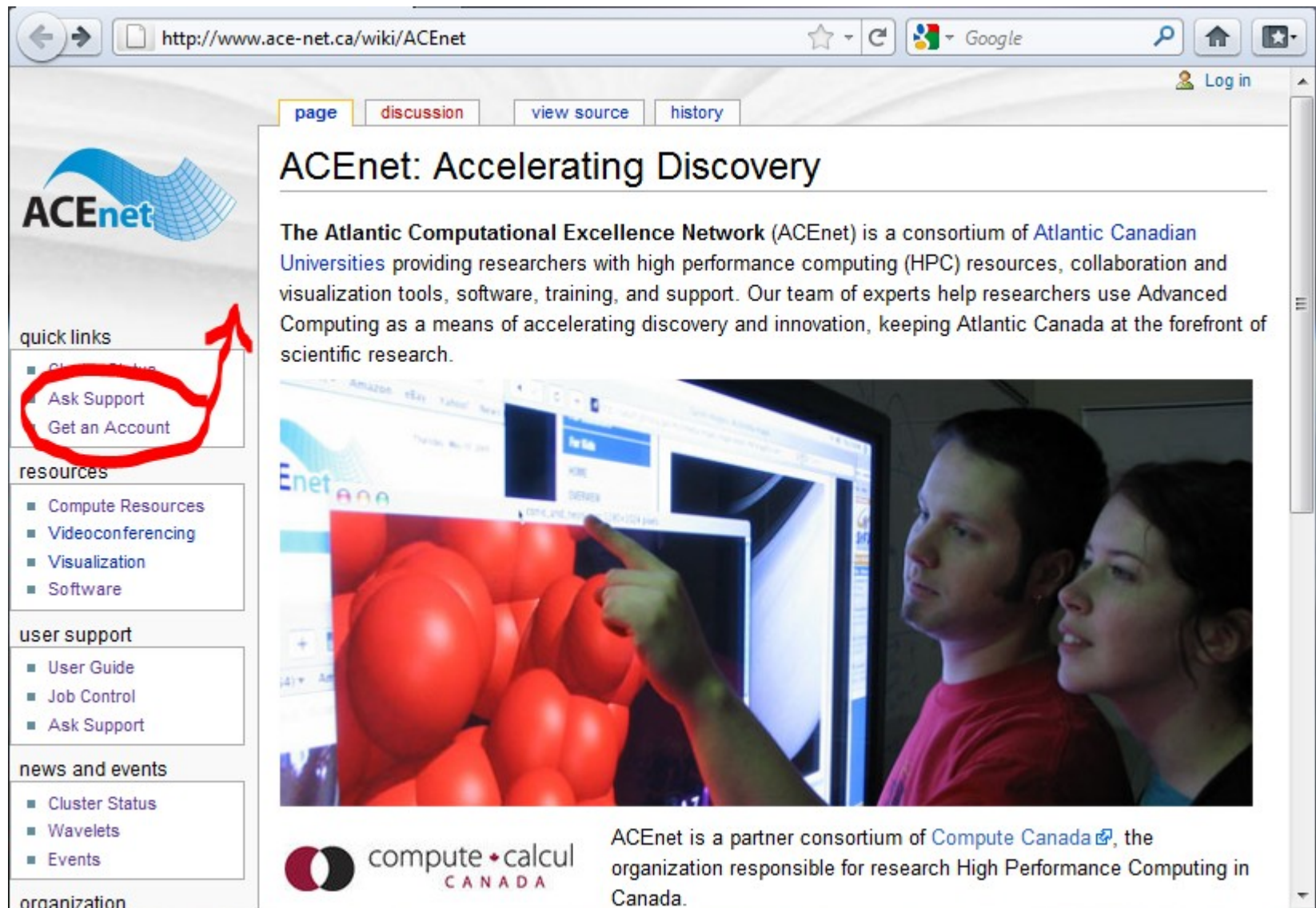
What's the cost?

- No money. It's paid for.
- Keep publishing.
 - and tell us about it, once a year
in conjunction with account renewal
- Share the resource.
 - about 250 projects
 - about 1000 user accounts
 - jobs must use dynamic resource manager
 - Sun Grid Engine

Organization and funding

- Consortium of Atlantic Cdn. universities
- Compute Canada is nat'l. umbrella org.
- Several \$million from
CFI + ACOA + provinces + “in kind”
- 10 technical staff @ 5 sites
- Research Directorate
Local User Groups ← user direction!
Resource Allocation Committee

Getting an account: <http://www.ace-net.ca/>

A screenshot of a web browser displaying the ACEnet website. The browser's address bar shows the URL "http://www.ace-net.ca/wiki/ACEnet". The page has a navigation bar with tabs for "page", "discussion", "view source", and "history". The main content area is titled "ACEnet: Accelerating Discovery" and contains a paragraph about the Atlantic Computational Excellence Network (ACEnet). Below the text is a photograph of two people, a man and a woman, looking at a computer screen. The man is pointing at the screen, which displays a 3D visualization of red spheres. The website's left sidebar contains several sections: "quick links" with a red circle around "Get an Account" and an arrow pointing to it, "resources" with links to "Compute Resources", "Videoconferencing", "Visualization", and "Software", "user support" with links to "User Guide", "Job Control", and "Ask Support", "news and events" with links to "Cluster Status", "Wavelets", and "Events", and "organization". At the bottom of the page, there is a logo for "compute + calcul CANADA" and a paragraph stating that ACEnet is a partner consortium of Compute Canada.

How Do I get Connected?



- Logging in
 - ssh *Secure Shell*
- Moving Data Around
 - sftp *Secure File Transfer Protocol*
 - scp *Secure Copy*
- both provided with **Mac OS X** or **Linux**
 - ssh -X username@brasdor.ace-net.ca
 - sftp username@placentia.ace-net.ca
- **Windows?** Use PuTTY and WinSCP
just google for them



Command-Line Interface



```
glooscap-head.cs.dal.ca - PuTTY
Using username "rdickson".
rdickson@glooscap.ace-net.ca's password:
Last login: Wed Mar 16 10:03:16 2011 from rdickson.ucis.dal.ca
Welcome to Glooscap.ace-net.ca - Halifax, Nova Scotia
*****

rdickson@glooscap: ~ $ /usr/local/R/bin/R

R version 2.8.0 (2008-10-20)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> █
```


Where to go for help

- ACEnet User Guide
 - http://www.ace-net.ca/wiki/User_Guide
- ACEnet User Wiki
 - <http://www.ace-net.ca/wiki/ACEnet>
- Research/user support personnel
 - email: support@ace-net.ca

Training Available

Seminars on demand (~ 1 hr each)

- Introduction to ACEnet
 - Introduction to Linux
 - Understanding the Grid Engine
 - Perl Programming
 - ...many more...
-
- Cross-Canada courses *via* videoconference
 - *e.g.* Parallel Programming course,
from HPCVL in Kingston, Ont., Feb 2011
 - Custom training
 - *your interest here*

What cluster should I use?



- Doesn't matter,
ONE ACCOUNT IS GOOD EVERYWHERE
- Multiple serial jobs? Brasdor best
- Shared memory parallel? Fundy best
- MPI? Mahone best, Placentia, Glooscap, Fundy good
- Some software has only local license
 - e.g. Matlab DCS, Gaussian only at Placentia
 - See <http://wiki.ace-net.ca> for availability, or contact your Computational Research Consultant

What can we do
for you?